SEGi UNIVERSITY

# BCC1313 – ALGORITHMS & COMPLEXITY

## Lab 2

**Instruction:**

Answer any **TWO (2)** of the questions below. Write C++/Java programs to solve the following recursion problems.

**1. Factorial Calculation:**

Write a program that calculates the factorial of a given number using recursion. The program should prompt the user for a positive integer and output the factorial of that number.

> The factorial of `n(n!)` is calculated as `n*(n-1)*(n-2)*...*1`.
> The base case occurs when `n == 1`.

**2. Recursive Fibonacci Sequence:**

Implement a recursive function to calculate the nth number in the Fibonacci sequence. The Fibonacci sequence is defined as:

```
F(0) = 0
F(1) = 1
F(n) = F(n-1) + F(n-2) for n > 1
```

Test the function by printing the Fibonacci numbers up to a given limit, entered by the user.

**3. Tower of Hanoi:**

Solves the Tower of Hanoi problem using recursion. The program should display the moves required to transfer all disks from the first rod to the last rod, following the rules of the Tower of Hanoi problem. Prompt the user for the number of disks.

**4. Sum of Array Elements:**

Write a recursive function that takes an array of integers and returns the sum of its elements. The function should have two parameters: the array itself and the size of the array.

- Task: Implement the recursive function and test it with different arrays.
- Hint: The base case will occur when the array has only one element.

**5. Reversing a String:**

Write a recursive function that takes a string as input and returns the string in reverse order. Do not use loops or built-in functions for reversing the string.

- Task: Test the function with various strings to check if it works as expected.
- Hint: The base case occurs when the string is empty or contains only one character.